

DLE602 Week 4 Interactive Python Tutorial Guide

Regularization, Overfitting, Underfitting and Generalisation

Purpose: This notebook lets you change model complexity and the regularization parameter λ (lambda) to see how training error, testing error, overfitting, underfitting and generalisation change in real time.

1. What you should achieve

- Understand why a model can perform very well on training data but poorly on unseen data.
- Explain underfitting, overfitting, good fit and generalisation using visual evidence.
- Observe how polynomial degree controls model complexity.
- Observe how λ controls the strength of L2 regularization.
- Compare training errors and testing error and decide which model is more reliable.
- Connect regularization to deep learning practice and assessment/project work.

2. How to run the notebook

1. Open DLE602_Week4_Regularization_Interactive_Tutorial.ipynb in Jupyter Notebook, JupyterLab, VS Code or Google Colab.
2. Run the notebook from the top. Do not skip cells, because later sections depend on earlier functions.
3. Read each short explanation before running the code cell below it.
4. When sliders appear, change one value at a time and observe the graph and printed error values.
5. Record at least three observations: one for underfitting, one for overfitting, and one for improved generalisation.

3. Important concepts in the program

Concept	Meaning	Clear Explanation
Training data	Examples used by the model to learn.	Like past exam questions used for practice.
Testing data	Unseen examples used to check generalisation.	Like a new exam question that tests real understanding.
Model complexity	How flexible or complicated the model is.	A low degree is too simple; a high degree can become too curvy and memorise noise.
Underfitting	Model is too simple and misses the real pattern.	The model has high bias and performs poorly even on training data.
Overfitting	Model memorises training data/noise and fails on new data.	Training error may be low, but testing error is high.
Generalisation	Ability to perform well on unseen data.	This is usually more important than perfect training accuracy.
Regularization	A technique that discourages unnecessary complexity.	It encourages the model to learn transferable patterns instead of memorising details.
λ (lambda)	Controls regularization strength.	Small λ = weak penalty; large λ = strong penalty; too large can cause underfitting.
L2 / Ridge	Penalises large weights and creates smoother models.	Useful when many features contribute a little and we want stable generalisation.

4. What to check in each notebook section

Notebook section	What to do	Expected outcome
Create dataset	Run the cell and inspect the noisy curved pattern.	You can see that real data has a pattern plus noise.
Train/test split	Compare training and testing points.	You can understand why unseen data is needed.
Polynomial degree slider	Move degree from low to high.	Low degree underfits; very high degree overfits.
λ slider	Keep degree high and increase λ gradually.	Curve becomes smoother; testing error may improve first.
Very large λ	Increase λ to a very high value.	Model becomes too simple and underfits.
Compare lambdas	Run comparison of several λ values.	You can observe that different λ values produce different model shapes.
Error vs λ plot	Inspect training and testing error curves.	Best λ is often not zero; lowest training error is not always best.
GenAI critique	Ask GenAI to explain regularization.	You can practice checking clarity, correctness and missing ideas.

6. Questions to answer while running the notebook

- Which degree clearly underfits the data? What evidence do you see?
- Which degree seems to be overfit? What happens to training and testing errors?
- When λ increases, what happens to the curve shape?
- Can regularization increase training error but still improve testing error?
- What happens when λ is extremely large?
- Which model would you deploy and why?

7. Key message to reinforce

The best model is not the one that memorises training data perfectly. The best model is the one that learns useful patterns and performs reliably on new, unseen data.

8. Post-class task

After completing the notebook, you should write a short reflection of 100–150 words answering:

- What is the difference between overfitting and generalisation?
- How did λ affect the training and testing errors?
- How could regularization be relevant to your Assessment 1 or future project work?